

Distinguishing between relations and functions



- 1 What is the name used to describe a graph where for some value of x , there exists two or more different values of y ?
- 2 Determine whether the following statements are true or false:
 - a When working with a function, substituting a certain value of x into the formula gives only 1 value of y for that value of x .
 - b A horizontal line can intersect the graph of a function at more than one point.
 - c A relation always passes the vertical line test.
 - d All functions are relations.
- 3 State whether each of the following statements are correct:

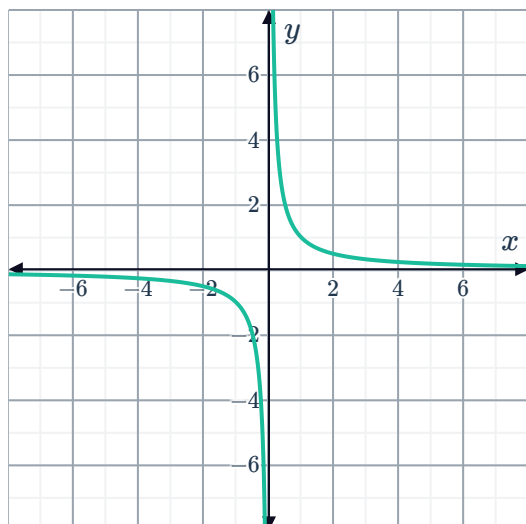
a Some relations are functions.	b All functions are relations.
c No functions are relations.	d No relations are functions.
- 4 State whether each of the following statements are correct:
 - a The graph of every non-vertical straight line is a function.
 - b There is no straight line graph that is a function.
 - c The graph of every straight line is a relation.
 - d The graph of every non-horizontal straight line is a function.

5 Determine whether each of the following

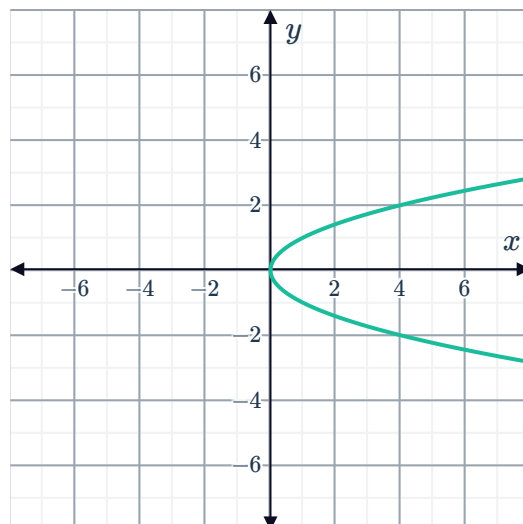


- a function,
- a relation,
- a relation and a function, or
- neither

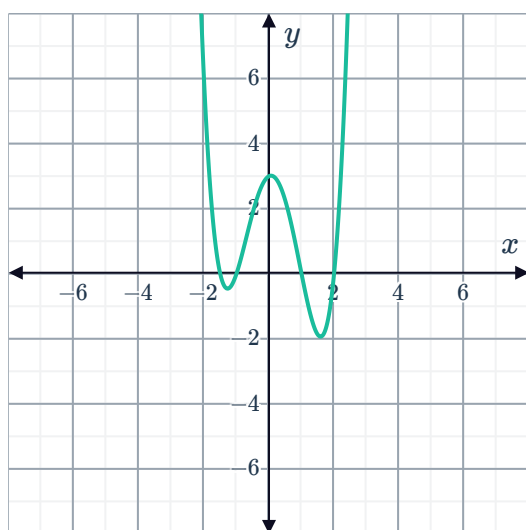
a



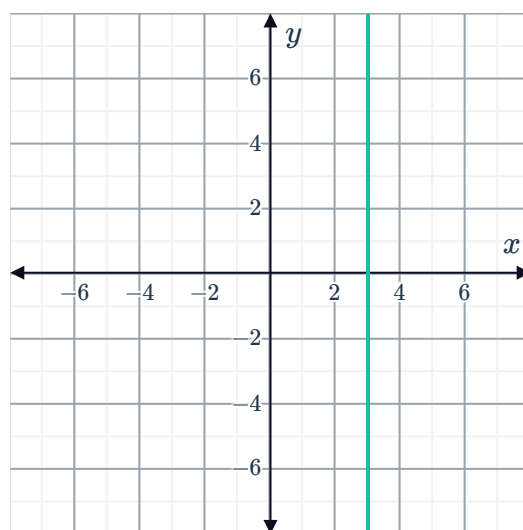
b



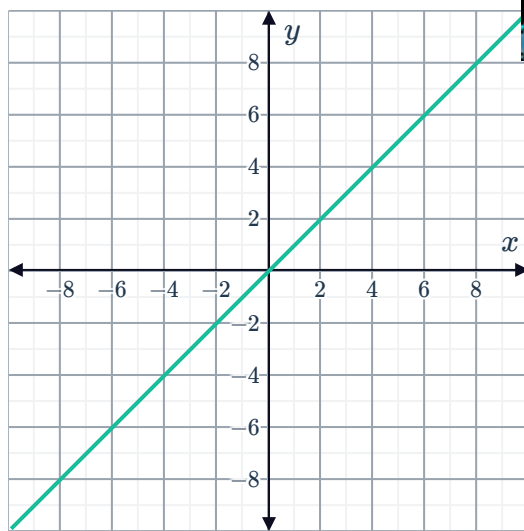
c



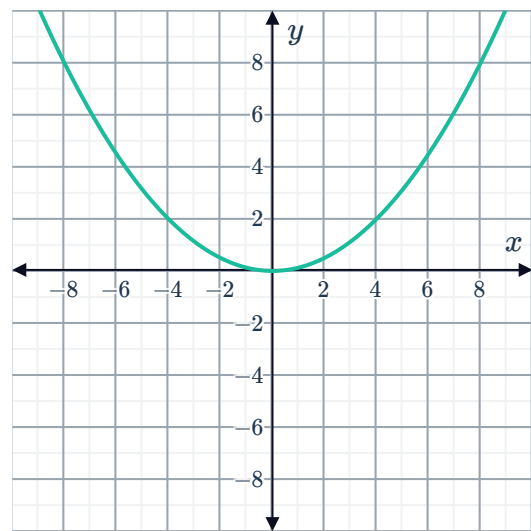
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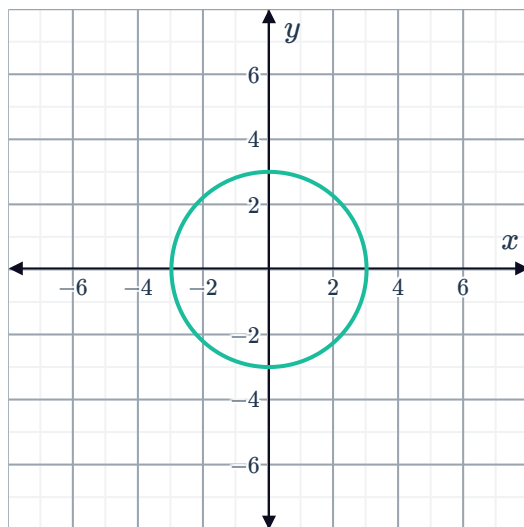
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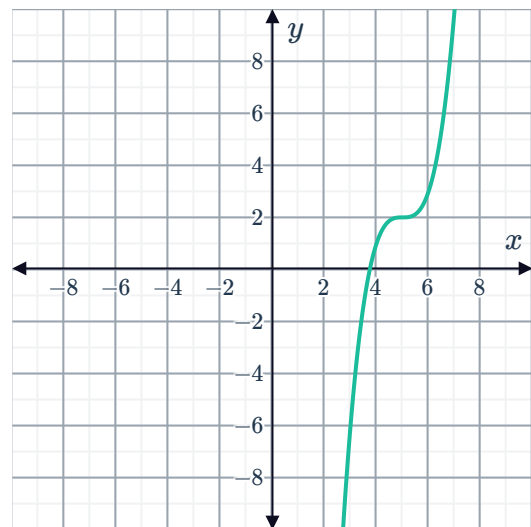
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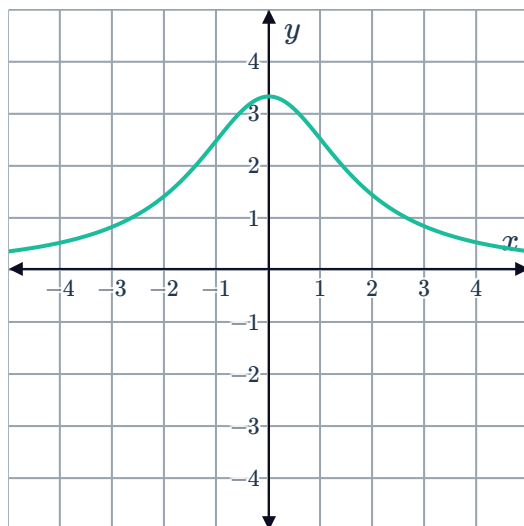
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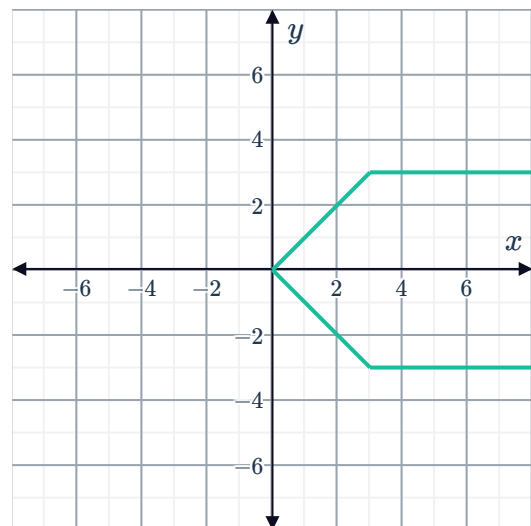
h



i



j

k $\{(2, 5), (7, -3), (5, 2), (-4, -9)\}$ l $\{(2, 5), (2, 7), (-3, -4), (-9, 13)\}$ m $(1, 5), (1, 1), (7, -2), (-5, -10)$ n $(1, 5), (7, -2), (-5, -10), (13, -13)$ o $(1, 5), (1, 7), (-2, -5), (-5, -10)$ p $y = 9x$ q $y = x^2 + 2$ r $y = \frac{3}{x}$

u $y = -x^3$



6 State whether the values in the tables below represent a function:

a

x	-4	-3	-2	-1	0	1	2	3	4
y	-4	-3	-2	-1	0	-1	-2	-3	-4

b

x	0	1	4	8	9	12	16	18	20
y	0	1	2	$\frac{2}{\sqrt{2}}$	3	$\frac{2}{\sqrt{3}}$	4	$\frac{3}{\sqrt{2}}$	$\frac{2}{\sqrt{5}}$

c

x	-9	-7	-6	-5	-3	-2	3	5	10
y	10	10	10	10	10	10	10	10	10

d

x	-3	-2	-2	0	1	1	2
y	8	4	3	7	2	1	0

7 Consider the points in the table.

x	-4	-3	-2	-1	0	1	2	3	4
y	4	3	2	1	0	1	2	3	4

a Plot the points on a number plane.

b Do they represent a function?

c Write down the coordinates of all the points that would need to be removed so that the remaining points form an increasing function.

8 Consider the points in the table:

x	-4	-3	0	-1	0	1	2	3	4
y	4	3	2	1	0	1	2	3	4

a Plot the points on a number plane.

b Do they represent a function?

9 A relation is defined as follows:



$y = 1$ if x is positive and $y = -1$ if x is 0 or negative.

a Complete the table for this relation.

x	-4	-3	-2	-1	0	1	2	3	4
y									

b Plot the points on a number plane.

c Do these values represent a function?

10 Consider these ordered pairs:

$\{(-9, -5), (-5, -10), (-5, -4), (-3, 7), (-2, -4), (-1, 1)\}$

a Plot the ordered pairs on a number plane.

b Which ordered pair would need to be removed from the set so that the remaining ordered pairs represent a function?

11 Write down a value of k such that the relation $\{(6, 2), (8, 5), (1, 7), (k, 4)\}$ does not represent a function.

12 Does there exist any values of k such that the relation $\{(2, 1), (7, k), (5, 8), (3, 9)\}$ does not represent a function?

Applications

13 CheapCalls Mobile charges \$1.10 a minute plus a connection fee of 70c for an international call.

a Complete the given table:

b Is this relation a function?

Call length (minutes)	International Call cost (dollars)
1	
2	
3	
4	
5	

- 14 A particular store is offering one free T-shirt for every two T-shirts purchased, where each T-shirt costs \$19.



- a Complete the table:
b Is this relation a function?

Number of T-shirts	Total Cost (dollars)
1	
2	
3	
4	
5	

- 15 A particular internet service provider charges \$40 a month on their 20 GB plan and charges \$10 for each additional 10 GB used.

- a Complete the given:
b Is this relation a function?

Total GB Used	Total Charge (dollars)
10	
20	
30	
40	
50	